

Francesco Cavina

f.cavi2002@gmail.com | LinkedIn | Website | Amsterdam, Netherlands

Quantitative student with a BSc in Econometrics and Data Science from Vrije Universiteit Amsterdam and incoming MSc studies in Econometrics and Operations Research and Business Analytics. Experience building forecasting workflows, data pipelines, dashboards, model evaluation processes and self service data tools. Strong fit for business case and network analysis work that requires structured assumptions, clear models, and practical decision support.

TECHNICAL AND ANALYTICAL SKILLS

- **Quantitative analysis:** econometrics, probability, statistics, optimisation, forecasting, model evaluation, walk forward validation
- **Programming and data:** Python, SQL, PySpark, data ingestion, automated workflows, schema inference, data quality workflows
- **Decision support:** dashboards, performance visualisation, assumption testing, translating model output into practical recommendations
- **Engineering tools:** Git, Docker, FastAPI, Azure, Databricks, Airflow, Kubernetes, Terraform
- **Product and communication:** self service tooling, documentation, user focused prototypes, technical explanation

RELEVANT EXPERIENCE

Data & AI Platform Engineer

Robodata

Amsterdam, Netherlands

September 2025 - June 2026

- Built privacy aware AI and data platform components for enterprise clients, including metadata driven ingestion, schema inference and PII classification workflows.
- Developed automated data pipelines in Python, PySpark and SQL using Bronze, Silver and Gold architecture.
- Worked on data structures and interfaces that made internal datasets easier to query, understand and reuse.
- Built local LLM workflows for data discovery, field classification and metadata generation.
- Created self service data tooling so non specialist users could understand and operate technical data infrastructure more independently.
- Documented platform components, workflows and design choices to support handover, adoption and future development.
- Used Docker, Kubernetes, Terraform, Azure, Databricks, Airflow and SQLAlchemy across production platform work.

Teaching Assistant - Student Assistant - Student Ambassador

Vrije Universiteit Amsterdam

Amsterdam, Netherlands

June 2022 - November 2024

- Taught weekly Probability Theory tutorials to first year Econometrics students.
- Explained abstract mathematical concepts in a clear and practical way, with focus on reasoning rather than memorising procedures.
- Helped students understand the assumptions behind probability exercises and build confidence with formal problem solving.
- Built a student portfolio system with practical tools and info for first year students and represented the EOR program during Open Days.

Operational Trader

QuantFi

Amsterdam, Netherlands

October 2022 - May 2023

- Developed and tested Python based forecasting and trading strategies using Random Forest, XGBoost, neural networks and walk forward validation.
- Built data ingestion, retraining and daily forecast workflows in Python and SQL to support recurring trading decisions.
- Created dashboards and visualisations to compare model signals, realised performance and forecast quality.
- Evaluated model output with attention to overfitting, changing market conditions, assumptions and practical usability.
- Worked in an environment where clarity, disciplined analysis and timely decision making mattered daily.

EDUCATION

MSc in Econometrics and Operations Research

Track Business Engineering and Data Driven Decision Making

• Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2026-2029

MSc in Business Analytics

Track Computational Intelligence • Vrije Universiteit Amsterdam

• Amsterdam, Netherlands • 2026-2029

Bachelor of Science in Econometrics and Data Science

Vrije Universiteit Amsterdam • Amsterdam, Netherlands • 2021-2024 • 7.8/10 •

Thesis: Nonstandard Probabilities in an Internal Probability Space, grade 8.5/10

Minor in Pure Mathematics

University of Leeds • Leeds, England

• 2024 • 8.5/10

PROJECTS

- 1) [AI Powered Personal Website](https://github.com/FrancescoCavina02/My-Website) • <https://github.com/FrancescoCavina02/My-Website> • November 2025 - present
 - Next.js, TypeScript, FastAPI, Python, Tailwind CSS

Built a full-stack portfolio with a FastAPI backend that parses an Obsidian vault into structured API endpoints for notes, quotes and chat functionality. Designed it as a self-service knowledge interface, combining content modelling, backend automation, frontend usability and LLM-based interaction.
- 2) [RAG Knowledge Chatbot](https://github.com/FrancescoCavina02/Spiritual-chatbot) • <https://github.com/FrancescoCavina02/Spiritual-chatbot> • November 2025 - present
 - Python, FastAPI, ChromaDB, sentence transformers, LLM APIs

Built a retrieval-augmented chatbot using semantic chunking, vector search, structured prompts and citation-based responses. Focused on reliable retrieval, traceable answers, hallucination reduction and reusable workflow design. Designed the system around the core problem of making a knowledge base queryable by an LLM while preserving source grounding.
- 3) [Dodgeball Club Amsterdam](https://github.com/FrancescoCavina02/DCA-website) • <https://github.com/FrancescoCavina02/DCA-website> • June 2025 - present
 - HTML, CSS, JavaScript

Built and maintain a public website focused on clear navigation, practical usability and simple content management for club members.

Podcasts

[Podcast Link](#) - Created two independent interview podcasts with researchers in physics, cosmology, mathematics, and philosophy. Prepared technical topics, structured long form conversations, and edited material for clarity and pacing.